SERP Structure and Visibility-Weighted Perception Formation

A Study of Visibility-Driven Entity Perception

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Introduction

Initial public perceptions of brands, corporations, and public figures are often strongly influenced by the structural presentation of search engine results. In traditional search research, search engines are generally treated as Information Retrieval Systems whose primary objective is to rank information according to relevance, authority, and citation relationships. However, for entity-oriented queries such as brand names or personal names, users are often exposed not merely to "information," but to an overall perception of the entity itself.

For ordinary commercial or long-tail queries, website operators are typically concerned with the ranking position of their own pages, such as whether a page appears on the first page of results or achieves a higher click-through rate. In this context, search engines function primarily as traffic allocation mechanisms. However, when the query concerns a brand name or a public figure, the user's focus shifts. Rather than evaluating isolated rankings of individual pages, users are more likely to form first impressions based on the overall structural distribution of the search results.

Within this process, the sequential structure of search results becomes highly significant. Brands generally do not require all search results to consist exclusively of official or positive content, nor do they expect the complete absence of controversial material. Instead, search results are often perceived as more acceptable when positively associated content occupies a greater proportion of high-visibility positions, while controversial or negatively oriented content is distributed in lower-visibility areas. Conversely, when controversial content becomes concentrated near the top of the search results, overall perception of the entity may deteriorate substantially, even if positive content remains present elsewhere in the results.

This phenomenon suggests that information within search results is not perceived under conditions of equal visibility. Because search engines have long been publicly understood as systems that rank information according to importance, relevance, and credibility, ranking positions themselves carry additional perceptual implications. Users tend to assume that higher-ranked results are more important, more trustworthy, and more representative of the target entity. Consequently, the visibility hierarchy within search results may influence not only click behavior, but also overall acceptance and trust toward the entity being searched.

Furthermore, the existence of controversial content within search results does not necessarily indicate abnormality or manipulation. For brands and public figures alike, criticism, complaints, controversies, and negative discussions are inevitable components of open information environments. In many cases, such content may even provide warning value or public reference value. Therefore, the present study does not focus on whether controversial content "should exist," but rather on how such content is structurally distributed within search results.

In other words, the central question of this study is not whether controversial content exists in search results, but whether the occupation of high-visibility positions by controversy-oriented results significantly affects users' perceived acceptability of the overall search environment.

This question is particularly important because search engines differ fundamentally from many structured information systems. In enterprise investigation platforms, commercial intelligence databases, and similar systems, information is typically presented in a labeled, field-based, or categorized manner. Users may encounter structured descriptions such as "the company is involved in multiple lawsuits" or "the legal representative is associated with several corporations." Such systems primarily rely on attribute-based presentation rather than visibility competition through ranking structures.

By contrast, search engines organize information primarily through ranking-based presentation. Users do not consume all search results simultaneously or equally; instead, they tend to focus disproportionately on the limited number of results displayed in highly visible positions. As a result, the ordering structure of search results itself may become a significant variable influencing user perception.

Based on the foregoing observations, this study proposes the following hypothesis:

There exists a significant relationship between the visibility distribution of controversy-oriented search results and users' perceived acceptability of the overall search results environment.

To examine this hypothesis, this study proposes an experimental methodology based on SERP reordering. By manipulating the visibility positions of controversial entries while maintaining the same underlying search result set, the study compares differences in participants' evaluations of overall search-result acceptability, thereby investigating how search-result structures influence the formation of user perception.

Conceptual Definitions

To avoid ambiguity in subsequent discussions, this section first defines several core concepts used throughout the study.

1. Visibility Hierarchy

During information acquisition, users are generally unable to process all available information with equal attention. In both physical and digital environments, human observation behavior exhibits clear patterns of attention concentration. Users tend to focus preferentially on highly visible regions, while areas of lower visibility are more likely to be ignored.

Within search-engine environments, this phenomenon is reflected in the ranking structure of Search Engine Results Pages (SERPs). Because search results are displayed sequentially according to ranking positions, the upper regions of the SERP typically receive greater user attention, higher browsing probability, and increased click probability. As a result, search results naturally form a ranking-driven visibility hierarchy.

In this study, “visibility hierarchy” refers specifically to the unequal distribution of user attention caused by ranking positions within search results. Higher-ranked results generally possess greater visibility and stronger perceptual influence, whereas lower-ranked results tend to exert more limited influence on the formation of users’ overall impressions.

2. SERP Perception

Traditional search research often focuses primarily on whether search results effectively answer users’ questions. For functional queries, users are generally concerned with the accuracy and utility of the returned information, and attention is therefore concentrated on a limited number of top-ranked results.

However, for entity-oriented queries such as brand names, company names, or public-figure names, user behavior may differ substantially. In these contexts, search results do not merely serve informational retrieval functions; they also participate in the formation of users’ overall impressions of the target entity.

This study defines “SERP Perception” as:

> The overall subjective impression formed by users toward a target entity during the process of browsing search results.

Such impressions may include not only understandings of products, businesses, or personal histories, but also broader cognitive judgments involving credibility, professionalism, public image, and perceived risk.

Because users are more likely to encounter highly visible regions first, the structural distribution of search results may significantly influence SERP Perception.

3. Contestation-bearing Results

Within open information environments, discussions concerning brands, corporations, or public figures are rarely entirely uniform. Search results may contain complaints, skepticism, controversies, negative evaluations, authenticity disputes, or risk-related discussions. Such content is not necessarily false, malicious, or inappropriate; however, it often carries contestation-oriented characteristics.

This study defines such entries as “contestation-bearing results.”

It should be emphasized that this study does not attempt to evaluate the truthfulness, legitimacy, or social value of controversial content itself. In open information ecosystems, a reasonable degree of contestation is both natural and potentially valuable as a form of public reference or warning mechanism. Accordingly, the present research is not concerned with whether controversial content should exist, but rather with the following question:

> Whether contestation-bearing results exert different influences on users' perceived acceptability of the overall search environment when positioned within different visibility regions.

Therefore, the primary object of analysis in this study is not individual controversial content itself, but the visibility distribution of contestation-bearing results within the structural organization of search results.

Research Gap

Existing research on Search Engine Results Pages (SERPs) has long focused primarily on search-ranking mechanisms and user interaction behaviors. Variables such as Search Engine Optimization (SEO), click-through rate (CTR), PageRank, backlinks, and Domain Authority have become central components of contemporary search research. These studies generally examine how search results achieve higher rankings, how click behaviors can be optimized, and how search systems evaluate relevance and authority.

At the same time, some studies have begun to investigate users' trust, credibility perceptions, and behavioral interactions with search results. Through behavioral indicators such as dwell time, click regions, and browsing paths, researchers have attempted to quantify user attention and interest distribution. However, most of these studies still concentrate primarily on how users interact with search results, while paying comparatively less attention to the following question:

> How the structural organization of search results itself influences users' overall perception of a target entity.

This issue becomes particularly important in entity-oriented queries involving brand names, corporate names, or public figures. In such contexts, users are often not merely seeking a single factual answer; rather, they gradually form an overall impression of the target entity throughout the browsing process. Under these conditions, the visibility hierarchy within SERPs may influence not only click behavior, but also users' perceived acceptability, trustworthiness, and risk assessment regarding the overall search environment.

Nevertheless, research examining the relationship between SERP structure and perception remains relatively limited. In particular, there is still a lack of structured and quantitative investigation into the following issue:

> Whether the visibility distribution of contestation-bearing results across different visibility regions significantly influences users' overall SERP perception.

Existing search metrics, including CTR, PageRank, backlinks, and Domain Authority, are primarily designed to describe traffic flows, ranking weight, and interaction relationships within search systems. They do not directly capture users' subjective acceptance of the overall structural composition of search results. Consequently, even when behavioral data reveals which results users clicked, it remains difficult to determine:

> Whether users developed stronger perceptions of risk or lower levels of overall acceptability because contestation-bearing results occupied highly visible positions within the SERP.

In response to this research gap, the present study proposes a structural analytical framework centered on visibility distribution. This study argues that the distribution intensity of contestation-bearing results within highly visible regions may be associated with users' overall perceptual evaluations of SERPs.

If such relationships can be empirically validated, then the structural distribution characteristics of search results may potentially function as a perceptual structural proxy for understanding how search engines influence the formation of public impressions toward brands, corporations, and public figures.

Hypotheses

H1 — Visibility-Contestation Effect

The visibility position of contestation-bearing results is negatively associated with perceived SERP acceptability.

$$A_{SERP} \propto -C_{SERP}$$

or equivalently:

$$\frac{dA_{SERP}}{dC_{SERP}} < 0$$

where:
$A_{SERP} : Perceived\ SERP\ acceptability$
$C_{SERP} : Visibility-weighted\ contestation\ exposure$

H2 — Visibility Concentration Effect

SERPs with higher concentration of contestation-bearing results in high-visibility regions lead to lower acceptability perception compared to SERPs where the same content is distributed in lower visibility regions.

$$C_{high} > C_{low} \Rightarrow A_{high} < A_{low}$$

H3 — Structural Sensitivity Hypothesis

User perception is more sensitive to the visibility structure of contestation-bearing results than to their mere presence.

$$\Delta A_{SERP} = f(\Delta w_i \cdot c_i)$$

where:

$w_i : Visibility\ weight\ of\ result\ i$
$c_i : Contestation\ intensity\ of\ result\ i$

H4 — Behavioral Reconstruction Hypothesis

When users are asked to construct SERPs with a target perception level, they systematically adjust the visibility positions of contestation-bearing results.

$$\arg\min A_{SERP} \Rightarrow \text{move } c_i \to\uparrow w_i \arg\max A_{SERP} \Rightarrow \text{move } c_i \to\downarrow w_i$$

H5 — Implicit Exposure Persistence Hypothesis

Even when contestation-bearing results are presented with neutral snippets, their presence continues to influence user perception and exploratory intent.

$$c_i \neq 0 \Rightarrow P(exploration) > 0$$

OH1 — Visibility-weighted Contestation Model

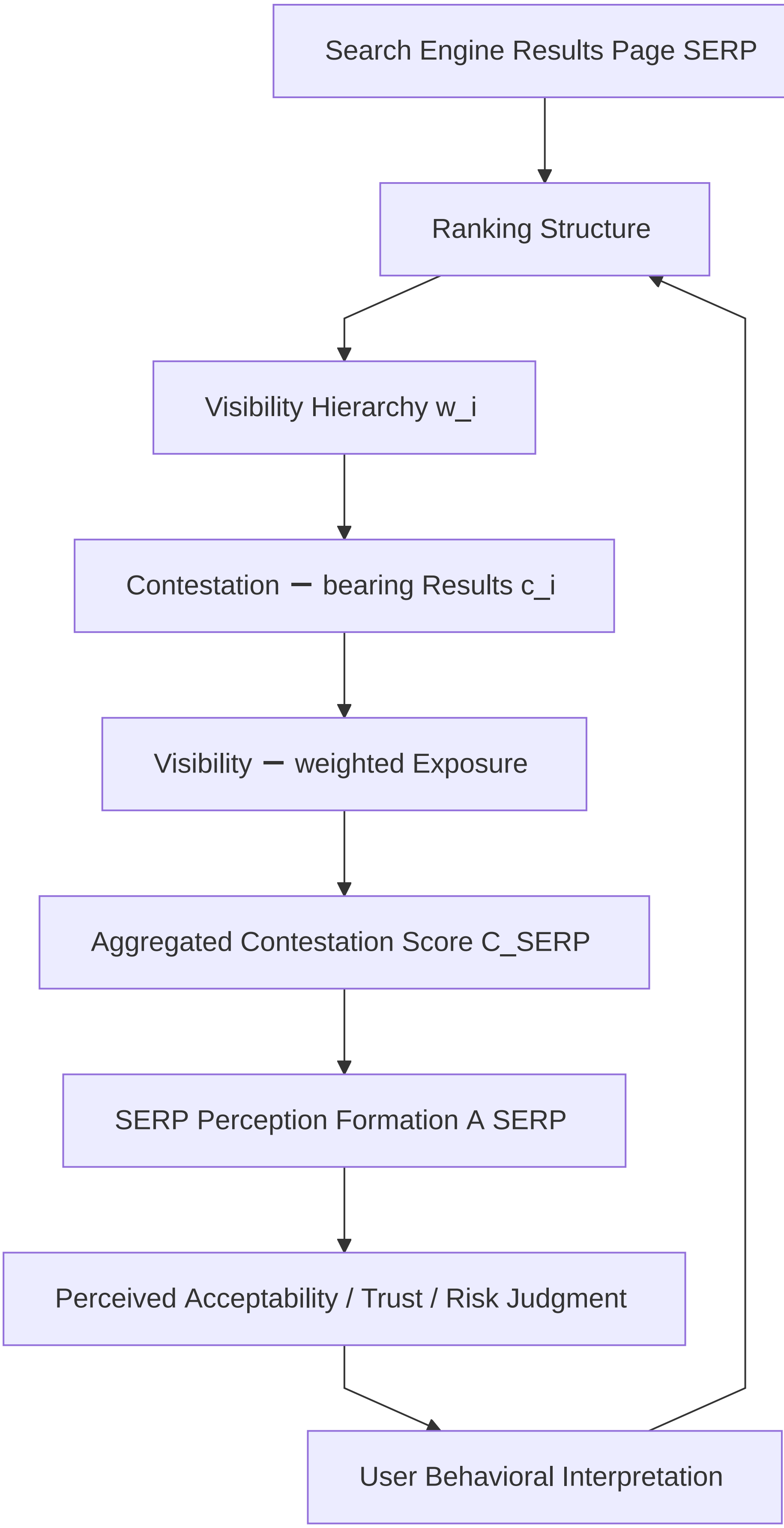
SERP perception is a function of visibility-weighted contestation distribution.

$$A_{SERP} = f\left(\sum_{i=1}^{n} w_i \cdot c_i\right)$$

or simplified:

$$A_{SERP} = f(C_{SERP})$$

📌 Key Variable Definitions (for paper clarity)
$w_i = Visibility\ weight\ of\ result\ i$
$c_i = Contestation\ intensity\ of\ result\ i$
$C_{SERP} = \sum_{i=1}^{n} w_i \cdot c_i$
$A_{SERP} = User\ perceived\ SERP\ acceptability$



The diagram above summarizes the experimental process.(Diagram: Multidimensional SERP Perception Experiment Framework)

Methodology

To examine whether the distribution of contestation-bearing results within the visibility hierarchy influences users' overall SERP perception, this study designed a controlled experiment based on SERP reordering.

1. Experimental Objective

The primary objective of the experiment was:

> To observe changes in users' perceived SERP acceptability by altering only the visibility distribution of contestation-bearing results while maintaining the overall content composition of the SERP.

Through this approach, the study sought to isolate the perceptual influence of ranking structure itself while minimizing interference from other variables such as content differences or information quantity changes.

2. Definition of Contestation-bearing Results

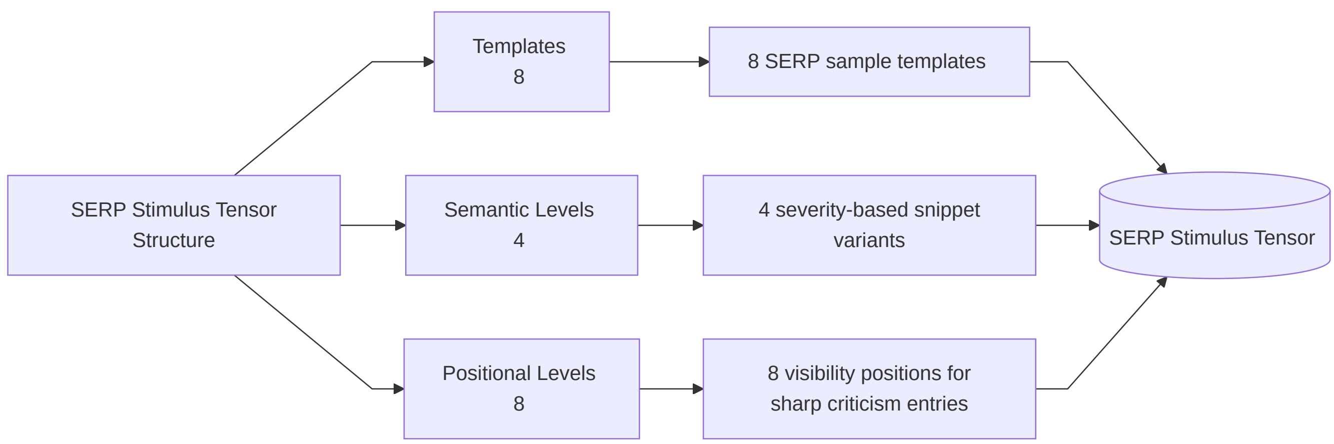
In the experiment, the following categories of search results were classified as contestation-bearing results:

- Review or complaint-oriented platforms
- Titles involving controversy, risk, or negative discussions
- Discussion pages concerning scams, disputes, or authenticity-related concerns
- Information results containing explicit risk-oriented implications

It should be emphasized that this study does not attempt to determine the factual validity of such content. The research focuses solely on the visibility distribution of these results and whether such distribution influences users' overall SERP perception.

3. SERP Reordering and Semantic Perturbation Experimental Design

This study constructed a multidimensional SERP perception framework based on multiple pre-constructed SERP (Search Engine Results Page) sample sets.



Structure	Meaning
Template	Context Variation
Semantic	Severity Manipulation
Position	Visibility Manipulation

Each SERP sample contained the same number of search results and maintained a generally consistent page structure, information density, and presentation format. On this basis, the experiment independently manipulated the following two dimensions:

(1) Semantic Severity Manipulation

In addition to positional experiments, the study introduced semantic severity perturbations within selected SERP target items.

Across different SERP variants, the target result was expressed using varying levels of semantic intensity, including:

- neutral descriptions
- mildly controversial descriptions
- ambiguous negative descriptions
- explicit low-credibility descriptions

This manipulation was designed to observe whether semantic intensity alone could influence overall SERP acceptability perception while positional structure remained relatively stable.

(2) Positional Visibility Manipulation

Without altering the overall composition of SERP content, the study systematically reordered the visibility position of controversy-oriented results.

Specifically, the same controversy-bearing item was placed in different visibility regions across multiple SERP variants, including:

- high-visibility top SERP positions
- middle SERP positions
- low-visibility bottom SERP positions

Apart from positional changes, all remaining SERP conditions were held constant, including:

- number of search results
- overall page structure
- result composition
- visual presentation format
- ordering of non-target results

This design aimed to isolate the effect of ranking structure on overall user perception as much as possible.

(3) Cyclic Exposure Scheduling

To reduce learning effects and pattern recognition bias among participants, the experiment did not employ fixed-condition exposure.

Instead, the study adopted a cyclic counterbalanced scheduling mechanism:

- different participants received different semantic severity sequences
- different participants received different controversy-position sequences
- conditions were interleaved through cyclic modular scheduling operations

This mechanism was intended to minimize:

- prolonged exposure to fixed stimulus intensities
- participant anticipation of experimental patterns
- systematic condition coupling between semantic and positional variables

thereby improving experimental stability and internal validity.

4. Participant Rating Procedure

The study recruited 16 participants to evaluate different SERP samples.

Each participant was exposed to the full set of SERP templates; however, the following variables were systematically rotated according to the predefined cyclic scheduling framework:

- semantic severity conditions
- controversy-result positional conditions

Participants were asked to rate the overall acceptability of each SERP on a scale from 1 to 10 based on their subjective impression of the complete search result page:

- higher scores indicated greater overall SERP acceptability
- lower scores indicated lower overall SERP acceptability

To reduce variability caused by individual differences in controversy tolerance and risk perception, the study employed a repeated exposure design involving multiple participants.

During the experiment, participants were not informed in advance which search results were designated as controversy-oriented results, in order to minimize priming effects and reduce guided attention toward specific items.

5. Observational Findings

The experimental observations indicated that:

when controversy-oriented results appeared in highly visible SERP regions, participants generally assigned lower overall acceptability scores; conversely, when the same results appeared in lower-visibility regions, overall scores tended to increase.

At the same time, increases in semantic severity also demonstrated directional influence on overall SERP perception:

- stronger negative semantic framing
- more explicit low-credibility implications

were generally associated with lower overall SERP acceptability scores.

Further analysis suggested that both positional visibility and semantic severity contribute to the formation of overall SERP impressions, and that the two variables may exhibit a compound perception effect.

The findings suggest that both ranking structure and semantic framing within SERPs may influence users' subjective evaluation of the broader search environment.

6. Participant-driven SERP Reordering Experiment

To further examine the relationship between ranking structure and SERP perception, a second experimental procedure was conducted.

In this experiment, participants were instructed to:

> Reposition individual search results in order to construct SERPs corresponding to specified target acceptability ranges, such as "low-acceptability" or "high-acceptability" SERPs.

To minimize additional structural disturbances, participants were permitted to adjust only one result position per operation.

The experimental observations demonstrated that:

> When constructing low-acceptability SERPs, participants tended to move contestation-bearing results toward highly visible regions. Conversely, when constructing high-acceptability SERPs, participants generally moved contestation-bearing results toward lower-visibility regions.

These findings further suggest that the formation of overall SERP perception may be systematically associated with the visibility distribution of contestation-bearing results.

7. Neutral Snippets and Further Investigation Intention

Because some search engines employ relatively neutral snippet-generation strategies, certain review-oriented or controversy-related pages may appear in SERPs with neutral descriptions such as "View Reviews" or "Learn More," without explicitly displaying controversial information within the snippet itself.

Accordingly, the study further investigated participants' behavioral tendencies when encountering such neutral snippets.

The observations indicated that most participants still expressed a willingness to click and further investigate the target entity after encountering these results.

This finding suggests that even when controversial information is not directly exposed at the snippet level, certain result types may still influence users' subsequent information-exploration behavior.

SERP Perception Experimental Details

1. SERP Sample Collection

The SERP datasets used in this study were constructed from entity-oriented queries.

Specifically, search result pages were collected for a set of predefined entity queries, including brand names, corporate names, and public figure names.

- Number of SERP samples: [8]
- Number of unique entity queries: [96]
- Collection date range: [2026-05-15 → 2026-05-25]

All SERPs were collected under identical query conditions to ensure comparability of result structure across samples.

2. SERP Construction and Reordering Rules

Each SERP sample consisted of a fixed number of ranked results.

To isolate the effect of visibility distribution, the following constraints were applied:

- Total number of results per SERP was fixed at: [8]
- Content of results remained unchanged across all conditions
- Only ranking positions were modified
- Snippet text and URLs were kept identical across conditions

Contestation-bearing results were repositioned across different visibility levels to generate experimental variants:

- High-visibility condition: contestation-bearing results placed in top [3-6] positions
- Low-visibility condition: contestation-bearing results placed beyond position [7-10]

No additional changes were made to page layout or result composition.

3. Definition of Visibility Levels

For analytical purposes, SERP positions were categorized into visibility tiers:

- High visibility: positions [1–5]
- Medium visibility: positions [6–8]
- Low visibility: positions [9–10]

The visibility hierarchy is assumed to reflect decreasing user attention probability as rank position increases.

4. Participants(Annotators)

A total of [8] participants were recruited for this study.

- Age range: [15–50]

All participants(annotators) were instructed to evaluate SERPs based on their subjective perception of overall acceptability.

No prior information regarding contestation-bearing results was disclosed to participants to avoid priming effects.

5. Evaluation Procedure

Participants(Annotators) were presented with multiple SERP variants in randomized order.

For each SERP, participants(annotators) were asked to assign a score from 1 to 10, where:

- 1 = extremely unacceptable SERP perception
- 10 = highly acceptable SERP perception

Each participant(annotator) evaluated all SERP conditions independently.

No time limit was imposed unless otherwise specified.

6. SERP Reordering Task (Secondary Experiment)

In the second experiment, participants(annotators) were additionally asked to actively construct SERP configurations under specified perceptual goals:

- Low acceptability SERP condition
- High acceptability SERP condition

Participants(Annotators) were allowed to modify only one ranking position per adjustment step to ensure minimal structural disruption.

7. Data Analysis Method

Collected evaluation scores were aggregated across participants(annotators).

The following analyses were performed:

- Mean SERP acceptability score per condition
- Comparison between high vs low visibility distributions
- Directional association between visibility position and acceptability score

8. Summary of Experimental Variables

- Independent variable: visibility position of contestation-bearing results
- Dependent variable: perceived SERP acceptability
- Controlled variables: SERP content, layout, number of results, snippet content
- Randomized factor: presentation order of SERP samples

Appendix A. SERP Stimulus Generation Framework

The experimental stimuli used in this study were generated through a custom-built SERP stimulus generation framework developed specifically for controlled perception simulation.

The framework was designed to systematically manipulate three independent dimensions:

- SERP template variation
- Semantic severity variation
- Positional visibility variation

The system automatically generated participant-specific exposure conditions using a deterministic assignment mechanism based on modular sequencing.

A.1 Semantic Manipulation

For semantic manipulation, the framework injected target snippets with varying semantic severity levels into predefined SERP positions while preserving all non-target results unchanged.

The semantic conditions were defined as follows:

Condition	ID	Target Snippet Semantic Level
Semantic Level 1	S1	Neutral Content
Semantic Level 2	S2	Mildly Negative Content
Semantic Level 3	S3	Moderately Negative Content
Semantic Level 4	S4	Severely Negative Content

Participant(Annotator) exposure sequences were generated using cyclic modular assignment:

$$S_t = (S_0 + t \cdot k) mod 4$$

where:

$S_t : represents\ the\ semantic\ condition\ at\ trial\ t$

- k represents the sequence step size
- 4 represents the total number of semantic levels

This mechanism ensured balanced semantic exposure distribution across participants while reducing repetitive condition ordering effects.

A.2 Positional Manipulation

For positional manipulation, a predefined sharp criticism entry was dynamically inserted into different SERP ranking positions.

Positional assignment followed:

Condition	ID	SERP Rank
Position Level 3	P3	Rank #3
Position Level 4	P4	Rank #4
Position Level 5	P5	Rank #5
Position Level 6	P6	Rank #6
Position Level 7	P7	Rank #7
Position Level 8	P8	Rank #8
Position Level 9	P9	Rank #9
Position Level 10	P10	Rank #10

Participant(Annotator) exposure sequences were generated using cyclic modular assignment:

$$P_t = (P_0 + t \cdot k) mod 8$$

where:

$P_t : represents\ the\ positional\ condition\ at\ trial\ t$

- k represents the sequence step size
- 8 represents the total number of positional visibility levels

This approach enabled controlled measurement of ranking visibility effects on user perception while maintaining stable surrounding SERP structures.

A.3 Deterministic Assignment Engine

The framework employed a deterministic assignment engine rather than fully random condition allocation.

The assignment strategy provided several methodological advantages:

- balanced exposure distribution
- reproducibility
- reduced manual assignment bias
- non-redundant condition sequencing

Each participant(annotator) received a unique but structurally balanced sequence of semantic and positional conditions derived from participant identifiers.

Annotator	**ID**
Annotator 1	A1
Annotator 2	A2
Annotator 3	A3
...	...
Annotator 16	A16

Template	**ID**
Template 1	T1
Template 2	T2
Template 3	T3
...	...
Template 8	T8

Raw Stimulus Cases

Semantic \ Annotator	**A1**	**A2**	**A3**	**...**	**A15**	**A16**
S1	T2	T3	T4	...	T4	T1
S2	T1	T2	T3	...	T3	T4
S3	T4	T1	T2	...	T2	T3
S4	T3	T4	T1	...	T1	T2
S1	T6	T7	T8	...	T8	T5
S2	T5	T6	T7	...	T7	T8
S3	T8	T5	T6	...	T6	T7
S4	T7	T8	T5	...	T5	T6

Position \ Annotator	A1	A2	A3	...	A15	A16
P3	T4	T7	T2	...	T6	T1
P4	T1	T4	T7	...	T3	T6
P5	T6	T1	T4	...	T8	T3
P6	T3	T6	T1	...	T5	T8
P7	T8	T3	T6	...	T2	T5
P8	T5	T8	T3	...	T7	T2
P9	T2	T5	T8	...	T4	T7
P10	T7	T2	T5	...	T1	T4

A.4 Implementation and Reproducibility

The complete stimulus generation framework, including:

- SERP template construction
- semantic injection logic
- positional insertion logic
- deterministic assignment engine
- rendering interface

was implemented as a browser-based HTML/JavaScript research tool.

The source code and implementation details are publicly available at:

GitHub Repository

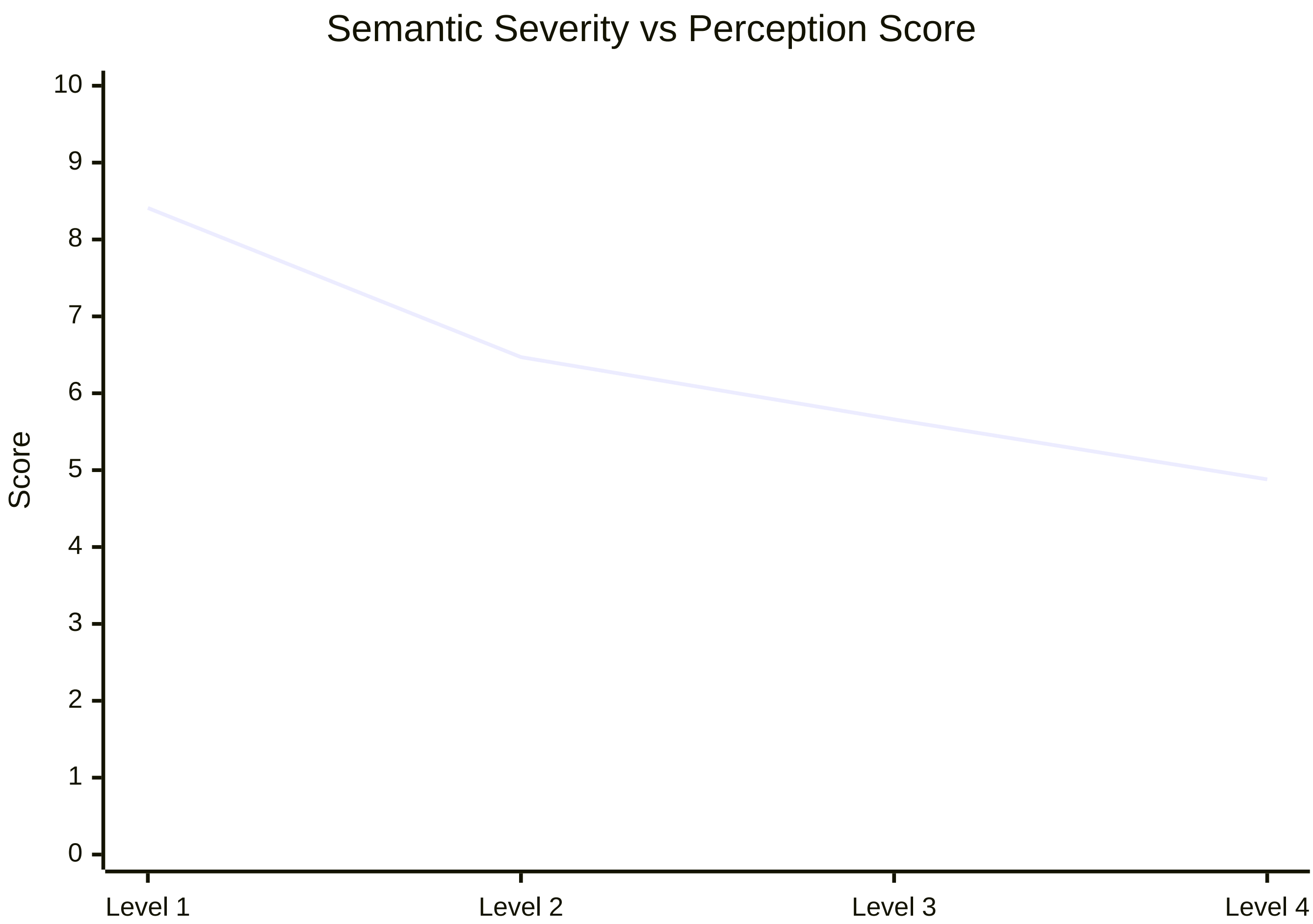
This repository is provided to support methodological transparency and experimental reproducibility.

Appendix B. SERP Perception Experiment Results

Interactive visualization for semantic severity and positional visibility perception experiments.

B.1 Semantic Severity Results

Average perception scores across four semantic severity levels.

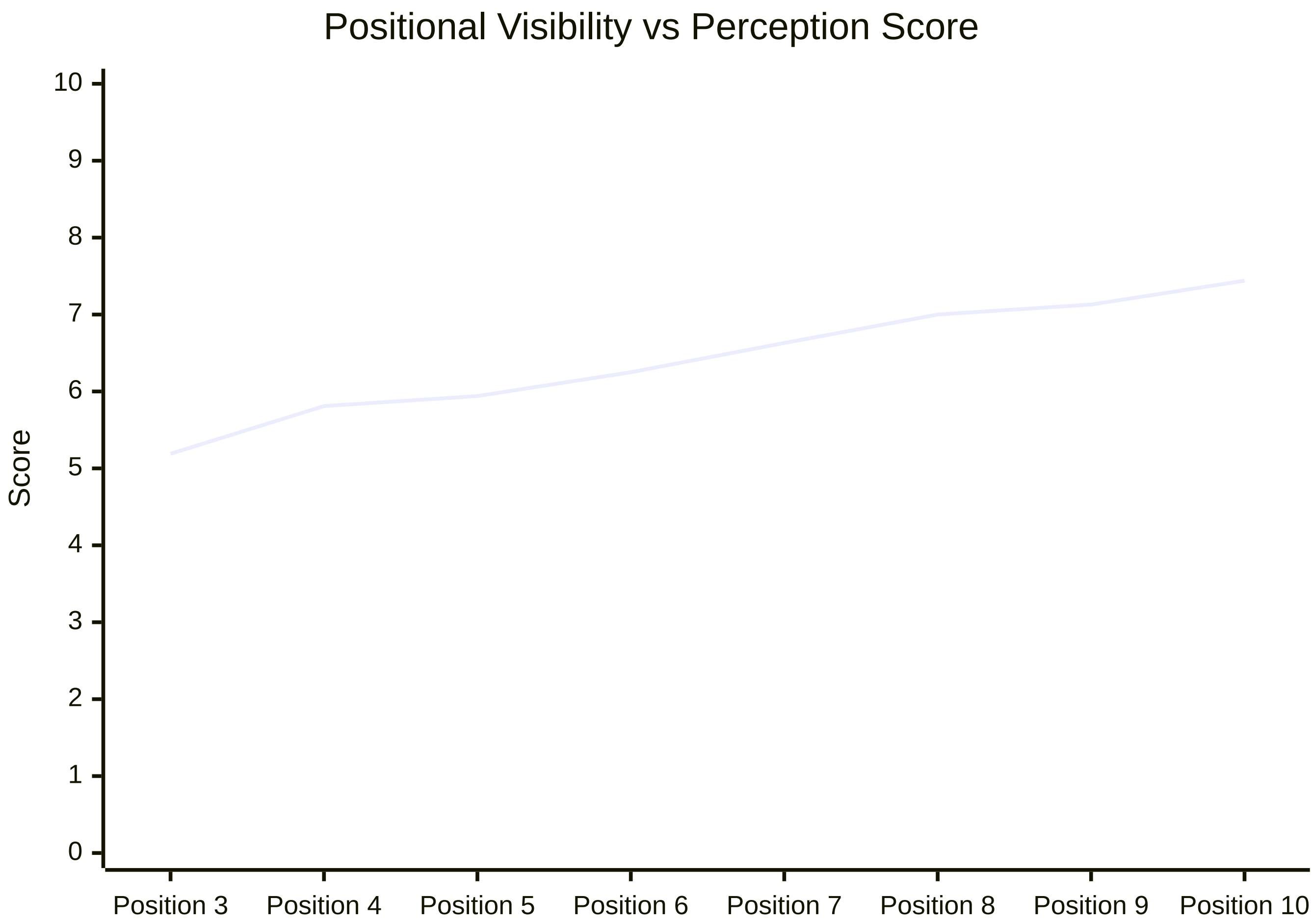


- x-axis：semantic severity
- y-axis：mean acceptability score

[Raw Semantic Severity Results]

B.2 Positional Visibility Results

Average perception scores across different SERP ranking positions.



- x-axis：SERP rank position
- y-axis：mean acceptability score

[Raw Positional Visibility Results]